Q77: THE $q = 0$ REVERSE SCHUR–WEYL CENTRALISER FAILS

RICK

From: Rick. To: Clio Vega (cc Robin Langer). Date: 2026-11-14.
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Proof record (local): `proofs/2026-11-13-q77-zero-schur-weyl-dc-fails.md` (PROVE 295); scripts `proofs/prove295/*.py`;
registry `registry/q77-zero-schur-weyl-dc.json` (this repository).

Grades.
- proved, written: the Theorem ($A \subsetneq B$ for $n \geq 2$, $N \geq 2n+1$). Internal grade: one internal cold read (Dream 294): PASS, 0 must-fix, 4 nits, one typo fixed. Not peer-reviewed.
- proved: for $N = 2n$, the "$a_n \cdot 1_\mu \in B$" half.
- computed (exact $\mathbb{Z}$ certificates): failure at $(3, 6)$ and $(4, 8)$; the "not in A" half at $N = 2n$ is checked exactly only at $n = 3, 4$.
- computed: the $n = 2$ values of $\dim A$ and $\dim B$ ($N \leq 10$).
- hunch: the two live classifications in §3.

1. Setup

$H = H_N(0)$ acts on $(k^n)^{\otimes N}$ by $v_w \cdot \pi_i = v_{ws_i}$ on ascents and $v_w \cdot \pi_i = v_w$ otherwise. Let A be the image of H, $S = \mathrm{End}_H = S_0(n, N)$ (owned by Deng–Yang / Jung–Oh), and $B = \mathrm{End}_S$. Then $A \subseteq B$.

2. Theorem

Theorem 1 (proved, written)**.** *For $n \geq 2$ and $N \geq 2n+1$, $A \subsetneq B$.*

- Let $h = \bar{\pi}_{w_0[1,n-1]} \cdot \bar{\pi}_{w_0[N-n+1,N-1]}$, where $\bar{\pi} = 1 - \pi$. For any set C of weights, $v \mapsto (v \cdot 1_C) \cdot h$ commutes with S.
- It is not in A whenever C separates two weights all of whose parts are ≥ 2.
- For $n = 2$ the witness is "$(1 - \pi_1)(1 - \pi_{N-1})$ on the weight space with $N - 2$ ones", for every $N \geq 5$.
- Proof idea: call a word *bad* if some letter $1, \ldots, n$ is missing, or the first occurrences of $1, \ldots, n$ are not in increasing order of position, or the last occurrences are not. Bad words are closed under the 0-Hecke moves, so they span a submodule Y. Being good is a necessary condition for reaching prefix = suffix = $12 \cdots n$ (no equivalence is claimed). Y is killed by h. Every cross-weight S-generator lands in Y. A local-idempotent argument shows the split operator is not in A.
- The proof is short and needs only standard facts.
- One-line picture: h maps $V^{\otimes N}$ into $\det \otimes V^{\otimes(N-2n)} \otimes \det$, where det is the $q = 0$ antisymmetriser line on n positions. S_0 cannot move weight through a det end, and H cannot tell two middle weights apart.

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3. Surprise

The census hunch "fails iff $N \geq 2n+1$" is wrong. $(3,6)$ also fails, by an exact $\mathbb{Z}$ certificate, and the mechanism there is different (non-split). The witness there is $a \cdot 1_\mu$ with $a = \bar{\pi}_{w_0[1,2]} \cdot \pi_3 \cdot \bar{\pi}_{w_0[4,5]}$ and $\mu = (1,2,3)$ or $\mu = (1,3,2)$.

- $(4,8)$ also fails. In general, at $N = 2n$ the element $a_n = \bar{\pi}_{w_0[1,n-1]}\, \pi_n\, \bar{\pi}_{w_0[n+1,2n-1]}$ gives $a_n \cdot 1_\mu \in B$. That half is proved; the "not in A" half is checked exactly only at $n = 3, 4$.
- Open: is $A = B$ at $(4,7)$? That decides between "$A = B \iff N \leq n+2$" and "$A = B \iff N \leq 2n-1$ $(n \geq 3)$".

4. The case $n = 2$: numbers

- $\dim A = \mathrm{Catalan}(N)$ (computed for $N \leq 10$; not proved).
- $\dim B = 43, 142, 493, 1768, 6475, 24052$ for $N = 5, \ldots, 10$. No closed form yet.

N	5	6	7	8	9	10
$\dim A$ (= Catalan(N), computed)	42	132	429	1430	4862	16796
$\dim B$	43	142	493	1768	6475	24052

5. Questions for Robin

(1) Is an MO 406152 answer wanted? It is the "naive Schur–Weyl duality for 0-Hecke" question.
(2) Is Deng–Yang §4 accessible to you? The novelty gate is still "not found".